

Strict Fractal-Wave Algebra (FWA) Axioms

Axiomatic Framework of Sequential Symmetry-Breaking

"Numbers are historical markers of sequential symmetry-breaking events."

"A perfectly balanced, symmetry-preserving baseline state."

1. Ontology of states

Axiom 1 (State space): There exists a set of states N , where each state $\Phi \in N$ represents a global symmetry configuration of a system.

Axiom 2 (Zero node): There exists a distinguished state $Z \in N$ such that Z is perfectly symmetric and undeformed. For all symmetry-breaking events AS , the effect of AS on Z defines the minimal non-trivial state:

$$AS(Z) \neq Z$$

Axiom 3 (Number as symmetry history): A number n is an equivalence class of finite sequences of symmetry-breaking events:

$$n \equiv [AS_1, AS_2, \dots, AS_k]$$

Two sequences are equivalent if they produce the same final state from Z .

2. Symmetry breaking and operations

Axiom 4 (Symmetry-breaking operator): Each event AS is a map:

$$AS: N \rightarrow N, \quad \Phi \rightarrow AS(\Phi)$$

such that $AS(\Phi) \neq \Phi$ unless AS is trivial.

Axiom 5 (Algebraic operation as transition): An algebraic operation $\cdot$ between numbers corresponds to a directed transition between states:

$$n_1 \cdot n_2 \iff \Phi_a \rightarrow \Phi_b$$

where $\Phi_a, \Phi_b \in N$ are states generated by the symmetry histories of n_1, n_2 .

Axiom 6 (Addition as translational deformation): Addition is defined as composition of translational symmetry-breaking:

$$n_1 + n_2 \equiv [AS^{(1)}_1, \dots, AS^{(1)}_{k_1}, AS^{(2)}_1, \dots, AS^{(2)}_{k_2}]$$

and corresponds to a net translational deformation of the underlying state:

$$\Phi_{n_1 + n_2} = T_{\text{trans}}(\Phi_{n_1}, \Phi_{n_2})$$

Axiom 7 (Multiplication as scaling deformation): Multiplication is defined as repeated or scaled application of symmetry-breaking:

$$n_1 \cdot n_2 \equiv \text{scale}([AS^{(1)}_1, \dots, AS^{(1)}_{k_1}], n_2)$$

and corresponds to a scaling deformation:

$$\Phi_{n_1 \cdot n_2} = T_{\text{scale}}(\Phi_{n_1}, \Phi_{n_2})$$

3. Dynamic arithmetic and collapse

Axiom 8 (Dynamic trajectory): Any computation is a trajectory in N :

$$\Phi_0 \rightarrow \Phi_1 \rightarrow \Phi_2 \rightarrow \dots \rightarrow \Phi_k$$

where each step is induced by a symmetry-breaking event AS_i .

Axiom 9 (Deformation operator): There exists a local deformation operator O such that:

$$O(\Phi, AS) = AS(\Phi)$$

and dynamic arithmetic is expressed as:

$$Da > (0+) > Pc$$

with Da the deformation amplitude and Pc the projected classical result.

Axiom 10 (Classical arithmetic as degenerate case): When local symmetry deformation tends to zero:

$$\lim_{AS \rightarrow 0} O(\Phi, AS) = \Phi$$

the dynamic process collapses to static classical arithmetic, and operations reduce to:

$$n_1 + n_2, \quad n_1 - n_2, \quad n_1 \cdot n_2, \quad n_1 / n_2$$

as emergent projections of the underlying wave dynamics.